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COSMOLOGICAL SELF-SIMILAR SHOCK WAVES AND GALAXY FORMATION

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ABSTRACT

A new family of similarity solutions is found for a point blast in a $\Omega = 1$ Friedmann cosmology (Einstein–de Sitter universe). The given solutions are valid several Hubble times after an explosion. Initially the shock wave formed satisfies the Sedov solution for adiabatic evolution in a constant medium, but it is modified by the cosmological expansion to form a self-similar thin dense shell which cools by adiabatic expansion and is unstable to gravitational fragmentation and collapse. The shock solution is applied to the explosive amplification model of galaxy formation of Ostriker and Cowie. It is shown that earlier than $1 + z \sim 9$, inverse Compton and free-free cooling prevent galactic-scale shocks from forming. At smaller redshifts, galactic shocks cool adiabatically to form cosmological dense-shelled shocks. Line cooling becomes important in the dense shells, cooling them to $\sim 10^4$ K and setting the Jeans mass to galactic scales. One generation of explosive amplification from an initial "seed" galaxy produces a shell of $\sim 3 \times 10^{11}\ M_\odot$ in galaxies, with radius and velocity dispersion typical of small groups. There may be time for a second generation (though no more); the resulting shell of $\sim 2 \times 10^{13}\ M_\odot$ has radius and velocity dispersion characteristic of small clusters of galaxies.

Subject headings: cosmology — galaxies: clusters of — galaxies: formation — relativity — shock waves

I. INTRODUCTION

The Sedov solution (Taylor 1950; Sedov 1959) for the adiabatic (postshock) evolution of a blast wave into a uniform medium is well known and frequently applied in astrophysics to supernova remnant evolution (see, e.g., Woltjer 1972). Other self-similar shock wave solutions have been found for adiabatic and isothermal evolution into media with power law density distributions (Sedov 1959; Solinger, Rappaport, and Buff 1975). Relativistic adiabatic similarity solutions have also been found (Blandford and McKee 1976). In spite of their diversity, however, these similarity solutions are all for explosions into external media which are constant in time.

Similarity solutions for adiabatic shock waves in cosmologically varying media (homogeneous in space but time-varying and self-gravitating) were cited by Schwarz, Ostriker, and Yahil (1975), but were not given in that paper. The solutions are presented here, and have recently been derived independently by Ikeuchi, Tomisaka, and Ostriker (1983). True similarity solutions exist only for Friedmann cosmologies with $\Omega = \rho/\rho_{\rm crit} = 0$ or 1, since only in these cases is the scale length a power law of time, though the general case of power law cosmological expansion is considered here. The basic equations and boundary conditions are given in § II, and the similarity solutions are derived and discussed in § III.

The cosmological adiabatic shock solution is applicable only for very high energy explosions in low-density media, since otherwise the shock radiatively cools and dissipates in less than a Hubble time, the time scale for cosmological effects to become important. But these conditions are exactly those encountered in the explosive amplification model of galaxy formation (Ostriker and Cowie 1981, hereafter OC; Ikeuchi 1981). OC expected that radiative cooling of galactic explosions would produce dense shells fragmenting to form new galaxies. In § IV we show that for moderate redshifts adiabatic cooling is more effective in forming dense shells (though radiative cooling governs shell fragmentation), and we discuss explosive amplification in terms of the similarity solutions given herein. The results are summarized in § V.

II. BASIC EQUATIONS

In this section we develop the notation and equations with which to describe cosmological similarity solutions. The fluid equations and boundary conditions are given first, and integrals of the motion are then derived from similarity arguments.

a) *Equations of Self-similar Flow*

Consider a point explosion in an expanding, self-gravitating, uniform medium with negligible pressure. The initial conditions of the explosion give the dimensional parameters t_i (cosmic time of the explosion, measured from the big bang), E (explosion energy), and ρ_i (cosmic density at t_i). From these and the independent variables t (cosmic time) and r, we can form a dimensionless time, and distance,

$$\tau \equiv \frac{t}{t_i}, \quad \lambda \equiv \frac{1}{\alpha}\frac{r}{t}\left(\frac{E}{\rho_i t_i^3}\right)^{-1/5} \tau^{1-\delta} \equiv \frac{r}{r_i}\tau^{-\delta}, \qquad (1)$$

where α and δ are constants to be determined. In the

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limit $\tau \gg 1$ we assume the shock propagation is given by

$$r_s(t) = \alpha \left(\frac{E t_i^{2-5\delta}}{\rho_i}\right)^{1/5} t^\delta = r_i \tau^\delta ; \quad (2)$$

$\lambda = 1$ defines the shock. Note that $\lambda = r/r_s$ is the radial coordinate normalized to unity at the shock.

A one-parameter similarity solution is sought for the velocity, density, pressure, and integral mass, of the form $y(r, t) = \zeta \tau^k Y(\lambda)$ for each dependent variable, where Y is a dimensionless function of λ, and ζ is a dimensional combination of t_i, r_i, and ρ_i. The most general form for self-similar motion is

$$v(r, t) = \frac{r_i}{t_i} \tau^a V(\lambda) , \quad \rho(r, t) = \rho_i \tau^b D(\lambda) ,$$

$$p(r, t) = \rho_i \frac{r_i^2}{t_i^2} \tau^c P(\lambda) , \quad m(r, t) = \tfrac{4}{3}\pi \rho_i r_i^3 \tau^d M(\lambda) , \quad (3)$$

with the constants a–d to be determined.

Now let us consider cosmology (cf. Weinberg 1972). Suppose $R(t) = R_i \tau^n$ is the scale length of the universe, with n constant. This is an exact solution of Einstein's field equations for a matter-dominated, homogeneous, and isotropic universe if $n = 0$ (Sedov case, no gravity and no Hubble flow), $n = 1$ (uniform expansion, no gravity, $\Omega = 0$), or $n = \frac{2}{3}$ (Einstein–de Sitter universe, $\Omega = 1$). Exact similarity solutions for shock waves in homogeneous cosmological media exist only in these cases, because in a similarity solution all nondimensional parameters without an explicit power law dependence on τ must be constant, and n is constant only if it equals 0, 1, or $\frac{2}{3}$. Then the preshock velocity and density in the laboratory frame (i.e., at the center of expansion) are

$$v = Hr , \quad \rho = \rho_i (R_i/R)^3 = \rho_i \tau^{-3n} , \quad (4)$$

where H is the Hubble constant. H and the density parameter Ω are related to n at any epoch by

$$H = \frac{\dot R}{R} = \frac{n}{t} , \quad \frac{\Omega}{2} = -\frac{\ddot R R}{\dot R^2} = \frac{1-n}{n} , \quad (5)$$

and the field equations relate these by

$$\frac{4}{3}\pi G\rho = \frac{\Omega}{2} H^2 = \frac{n(1-n)}{t^2} . \quad (6)$$

The fluid equations in Eulerian form, for a spherically symmetric, adiabatic, self-gravitating gas (neglecting magnetic fields, viscosity, thermal conduction, and radiation) are given by

$$\frac{\partial \rho}{\partial t} + \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 \rho v) = 0 , \quad (7a)$$

$$\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial r} = -\frac{1}{\rho}\frac{\partial p}{\partial r} - \frac{Gm}{r^2} , \quad (7b)$$

$$\left(\frac{\partial}{\partial t} + v\frac{\partial}{\partial r}\right) p\rho^{-\gamma} = 0 , \quad (7c)$$

$$\frac{\partial m}{\partial r} = 4\pi r^2 \rho . \quad (7d)$$

These are respectively the continuity, Euler, adiabatic, and mass equations. Here γ denotes the ratio of specific heats C_P/C_V (assumed constant). For $r > r_s$, equations (4), (5), and (7) give the Hubble flow boundary conditions ahead of the shock:

$$v = n\frac{r}{t} = n\frac{r_i}{t_i}\tau^{\delta-1}\lambda , \quad \rho = \rho_i \tau^{-3n} ,$$

$$m = \tfrac{4}{3}\pi\rho r^3 = \tfrac{4}{3}\pi\rho_i r_i^3 \tau^{3(\delta-n)}\lambda^3 . \quad (8)$$

We further require $p(r > r_s, t) = 0$. (The addition of external pressure adds another dimensional parameter, making a similarity solution impossible. The assumption made here is valid for a strong shock, whose Mach number greatly exceeds 1.) For $\lambda > 1$ (subscript 1 to denote preshock medium), equation (8) implies

$$V_1 = n\lambda , \quad D_1 = 1 , \quad P_1 = 0 , \quad M_1 = \lambda^3 . \quad (9)$$

Equations (7) and (8) also give the constants in equation (3):

$$a = \delta - 1 , \quad b = -3n , \quad c = 2(\delta - 1) - 3n ,$$

$$d = 3(\delta - n) . \quad (10)$$

The fluid equations (7) may be written in nondimensional form using equations (1), (3), (6), and (10). The result is

$$(V - \delta\lambda)D' + DV' + \frac{2DV}{\lambda} - 3nD = 0 , \quad (11a)$$

$$(V - \delta\lambda)V' - (1-\delta)V = -\frac{P'}{D} - n(1-n)\frac{M}{\lambda^2} , \quad (11b)$$

$$(V - \delta\lambda)\left(\frac{P'}{P} - \gamma\frac{D'}{D}\right) = 2(1-\delta) - 3n(\gamma - 1) , \quad (11c)$$

$$M' = 3\lambda^2 D , \quad (11d)$$

where a prime $\equiv d/d\lambda$. The self-similar nature of the solution has reduced the partial differential fluid equations into ordinary differential equations.

To integrate equations (11) we need the solution immediately postshock (denoted by subscript 2), since the fluid variables are discontinuous across the shock. They are given by the usual shock jump conditions derived from continuity of mass, momentum, and energy fluxes in the shock reference frame (cf. Spitzer 1978, § 10.2). When solved for the postshock values and written in dimensionless form, they become:

$$V_2 - \delta = \left(\frac{2\gamma}{\gamma+1}\right)\frac{P_1}{D_1(V_1 - \delta)} + \left(\frac{\gamma-1}{\gamma+1}\right)(V_1 - \delta) , \quad (12a)$$

$$D_2 = \left(\frac{\gamma+1}{\gamma-1}\right)\frac{D_1(V_1-\delta)^2}{[2\gamma/(\gamma-1)]P_1/D_1 + (V_1-\delta)^2} , \quad (12b)$$

$$P_2 = \frac{2}{\gamma+1}D_1(V_1-\delta)^2 - \left(\frac{\gamma-1}{\gamma+1}\right)P_1 , \quad (12c)$$

$$M_2 = M_1 , \quad (12d)$$

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where δ is the dimensionless shock velocity (cf. eq. [2]):

$$v_s = \frac{dr_s}{dt} = \delta \frac{r_s}{t} = \delta \frac{r_i}{t_i} \tau^{\delta - 1} . \quad (13)$$

Substituting the preshock boundary conditions (eqs. [9]) gives

$$V_2 - \delta = \frac{\gamma - 1}{\gamma + 1}(n - \delta) , \quad D_2 = \frac{\gamma + 1}{\gamma - 1} ,$$

$$P_2 = \frac{2}{\gamma + 1}(n - \delta)^2 , \quad M_2 = 1 . \quad (14)$$

b) *Integrals of the Motion*

Integrals of the motion are necessary to specify α and δ and to check the integration. Integrals expressing conservation of mass and energy are conveniently found using the similarity property of the fluid motion (cf. Sedov 1959; Landau and Lifshitz 1959, § 99). Consider a sphere of radius r, expanding such that λ = constant; the radial velocity of points on this surface is λv_s. Similarity requires that the mass contained in this sphere remain a constant fraction of the total mass bounded by the shock as it evolves. In a time dt the mass in the sphere increases by $4\pi r^2 \rho(\lambda v_s - v)dt$, while the total mass interior to the shock ($\lambda = 1$) increases by $4\pi r_s^2 \rho_1 (v_s - v_1)dt$. Thus we must have

$$\frac{4\pi r^2 \rho(\lambda v_s - v)dt}{4\pi r_s^2 \rho_1 (v_s - v_1)dt} = \frac{m}{(4/3)\pi r_s^3 \rho_1} , \quad (15)$$

which in dimensionless form becomes

$$M = \lambda^2 D \frac{(V - \delta\lambda)}{n - \delta} . \quad (16)$$

This result may also be obtained from direct integration of equations (11a) and (11d).

To acquire a second integral, we note that the change in energy of the gas enclosed by the shock equals the energy of the swept-up gas. For an adiabatic self-similar shock this must vanish, because otherwise the total energy enclosed by the shock, divided by the initial explosion energy, would not be a constant. (This does not exclude self-similar detonation waves, which have $E = 0$ at $t = t_i$.) Similarity requires this energy balance to hold for any sphere λ = constant. In a time dt the amount of energy $4\pi r^2 v dt[p + \frac{1}{2}\rho v^2 + p/(\gamma - 1) - Gm\rho/r]$ leaves the sphere of radius r. But as the sphere expands, the energy acquired in the extra volume is $4\pi r^2 \lambda v_s dt \times [\frac{1}{2}\rho v^2 + p/(\gamma - 1) - Gm\rho/r]$. Equating these gives

$$pv + (v - \lambda v_s)\left(\frac{1}{2}\rho v^2 + \frac{p}{\gamma - 1} - \frac{Gm\rho}{r}\right) = 0 , \quad (17)$$

which in dimensionless form becomes

$$PV + (V - \delta\lambda)\left(\frac{1}{2} DV^2 + \frac{P}{\gamma - 1} - n(1 - n)\frac{MD}{\lambda}\right) = 0 . \quad (18)$$

The shock velocity δ is now specified by differentiating equation (18) and substituting equations (11), (16), and (18), giving

$$(3n + 2 - 5\delta)\left(\frac{1}{2} DV^2 + \frac{P}{\gamma - 1} - n(1 - n)\frac{MD}{\lambda}\right) + n(1 - n)(2 - 3n)\frac{MD}{\lambda} = 0 . \quad (19)$$

For $n = 0$, 1, or $\frac{2}{3}$, this will be satisfied if

$$\delta = \tfrac{1}{5}(3n + 2) . \quad (20)$$

This result may also be shown heuristically from the results of the Sedov solution, for which $r_s \propto (Et^2/\rho)^{1/5} \propto t^{\delta}$, noting that cosmologically $\rho \propto t^{-3n}$.

A third integral of motion may be derived by exploiting the adiabatic nature of the postshock flow, using the specific entropy as a Lagrangian variable.[1] The specific entropy of a given fluid element is $S = C_V \ln q$, where

$$q = e^{S/C_V} = p\rho^{-\gamma} = p_2 \rho_2{}^{-\gamma} , \quad (21)$$

where p_2 and ρ_2 are the values the fluid element had just after it crossed the shock. Since it is constant along an adiabat, q may be used as an independent variable by labeling fluid elements with the value of q they had after they were shocked. But for spherical symmetry we may also use the integrated mass m as a Lagrangian variable, labeling each fluid shell by the mass interior to it. For different fluid shells, both q and m are functions of the dimensionless time τ_s when a given fluid shell was shocked. Using equations (3) and (10) to eliminate τ_s between q and m gives

$$q = q_0 (m/m_0)^{(\gamma n - \delta)/(\delta - n)} , \quad (22)$$

where q_0 and m_0 are dimensional constants depending on γ, E, t_i, and ρ_i. Substituting this in equation (21) gives the integral

$$p\rho^{-\gamma} m^{(\gamma n - \delta)/(n - \delta)} = \text{constant} . \quad (23)$$

This result is valid for all values of τ_s (along all adiabats), and hence for all values of τ and $\lambda \leq 1$. Nondimensionalizing and using equation (20) gives

$$PD^{-\gamma} M^{(\gamma n - \delta)/(n - \delta)} = \text{constant} . \quad (24)$$

This integral may also be obtained by direct integration of equations (11a) and (11c). Note that the fact that equation (23) must be independent of τ provides an independent derivation of equation (20).

Global integrals may be obtained using conservation of total energy and the virial theorem. Integrating over the volume enclosed by the shock and nondimensionalizing, energy conservation gives for α (defined in eq. [2])

$$\alpha = [4\pi(t + u + w)]^{-1/5} , \quad (25)$$

[1] I am indebted to Dr. J. P. Ostriker for pointing this out.

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where t, u, and w are the nondimensional total kinetic, thermal, and gravitational energies:

$$t = \int_0^1 \tfrac{1}{2} D V^2 \lambda^2 d\lambda , \quad u = \int_0^1 \frac{P}{\gamma - 1} \lambda^2 d\lambda ,$$

$$w = -n(1-n) \int_0^1 \frac{MD}{\lambda} \lambda^2 d\lambda . \tag{26}$$

The virial theorem is obtained by multiplying the Euler equation by ρr and integrating over the shock-enclosed volume, using the continuity equation and equation (14) for the surface terms obtained by using Gauss's theorem. The nondimensional result is

$$\tfrac{1}{2}\ddot{I} = \int_0^1 \lambda D V \lambda^2 d\lambda = 2t + 3(\gamma - 1)u + w - n(n - \delta) . \tag{27}$$

III. COSMOLOGICAL SIMILARITY SOLUTIONS

Solutions are now given of the nondimensional fluid equations (11), with boundary conditions set at $\lambda = 1$ by equations (14). The $n = 0$ noncosmological Sedov solution is already well known (Sedov 1959; Zel'dovich and Raizer 1966). The trivial $n = 1$ ($\Omega \ll 1$) case is discussed below first, and then accurate numerical solutions are given for the $n = \frac{2}{3}$ ($\Omega = 1$) case. The cosmological solutions are only valid for $\tau \gg 1$, and the development from a Sedov solution for $\tau \ll 1$ is discussed. Finally, the $n = 2/3$, $\gamma = 5/3$ solution is argued to be gravitationally unstable.

a) *Open Universe* ($\Omega = 0$)

For $n = 1$ (negligible gravity, uniform expansion), only trivial solutions of equations (11) exist, with $V = \lambda$, $P = 0$, and D arbitrary. (D is given by non–self-similar initial conditions.) The "shock" comoves with the expanding medium ($r_s \propto t$), so no matter passes through it and no shock heating occurs. (Thus there is no shock in the usual sense, although there is a contact discontinuity, with the density discontinuous at $\lambda = 1$.) This solution is the limiting case after many Hubble times of a point blast in a $\Omega \ll 1$ Friedmann cosmology. Before reaching this self-similar solution an initial Sedov blast wave decelerates until it is comoving with the Hubble flow. The (nondimensional) density distribution at that time is frozen in, and adiabatic expansion cools the interior to $p = 0$.

b) *Einstein–de Sitter Universe* ($\Omega = 1$)

In the $n = \frac{2}{3}$ (Einstein–de Sitter–Sedov) case, equation (20) shows that the shock propagates as $r_s \propto t^{4/5}$. Cosmological effects modify the solution substantially from the Sedov solution. Mathematical singularities appear, and it is necessary to factor out the singular behavior before numerical solution can be obtained. (An analytic solution was not found for this case.)

The independent variable is first changed from λ to

$$x \equiv \delta\lambda - V . \tag{28}$$

This is necessary because, as will be seen, the solutions do not extend all the way to $\lambda = 0$, but only to $x = 0$ (where $\lambda = \lambda_0 > 0$ and $V \neq 0$). It is easy to see that $x \geq 0$. Suppose a fluid element has dimensionless velocity $V = \delta\lambda$. As it evolves it then remains at constant λ (constant fraction of the shock radius). But if $V > \delta\lambda$, the fluid element moves toward the shock (increasing λ) and the profile changes with time, destroying the self-similar nature. Since x is nonnegative and numerical integrations show it is monotone increasing with λ, it is a suitable independent variable.

Expansion of equations (11) for $n = \frac{2}{3}$ to lowest order in x shows a power law behavior of the solutions near $x = 0$ for $\gamma > 7/6$. (The solution for $\gamma \leq 7/6$ has not been found.) This behavior is factored out in order to allow accurate numerical solution:

$$D = \tilde{D}(x) x^{s-1} , \quad P = \tilde{P}(x) x^s , \quad M = \tilde{M}(x) x^s , \tag{29}$$

with

$$s = \frac{\gamma}{6\gamma - 7} . \tag{30}$$

V (and hence λ), $\tilde{D}$, $\tilde{P}$, and $\tilde{M}$ approach constant positive values V_0, D_0, P_0, and M_0 at $x = 0$ ($\lambda = \lambda_0$) which depend on boundary conditions at $\lambda = 1$ and must be determined numerically.

Equations (11), with equations (28) and (29) substituted, were integrated with 1000 points for $\gamma = 5/3$, 3/2, 7/5, and 4/3 using a fifth and sixth order Runge-Kutta ordinary differential equation solver. At each point the integrals (16), (18), and (24) were checked and had relative errors less than 10^{-11} (except at $x = 0$, where extrapolation of a power law fit provided slightly worse accuracy), so the solutions are believed to be highly accurate.

The solutions for $\gamma = 5/3$ and 7/5 are shown in Figures 1 and 2 and are presented in tabular form in Tables 1 and 2. For all values of $\gamma > 7/6$ the integrated mass is zero at $\lambda = \lambda_0 > 0$. Thus the matter swept up by the shock is confined to a shell of thickness $1 - \lambda_0 \approx 0.03$, interior to which is void. Intuitively this occurs because in the absence of a shock the fluid is already expanding due to the Hubble recession, and the additional kinetic energy of the explosion is enough so that the fluid comoves with the shock; hence $V = \delta\lambda$ at $\lambda = \lambda_0$. The pressure also goes to zero at $\lambda = \lambda_0$ as the fluid undergoes adiabatic decompression. As a result, the fluid flow in the laboratory frame is highly supersonic through the shell; in the shock frame the flow is subsonic for $\lambda > 0.969$ (for $\gamma = 5/3$ and 7/5) and supersonic interior to this. For $\gamma > 7/5$, equations (29) and (30) show that the density becomes singular at $\lambda = \lambda_0$, although slowly enough that the total mass enclosed by the shock is finite. This density singularity is unstable to further gravitational condensation and, as we shall see, may play an important role in galaxy formation.

Several constants associated with the Einstein–de Sitter–Sedov solutions are shown in Table 3 for $\gamma = 5/3$, 3/2, 7/5, and 4/3. The integrals in equation (26) were

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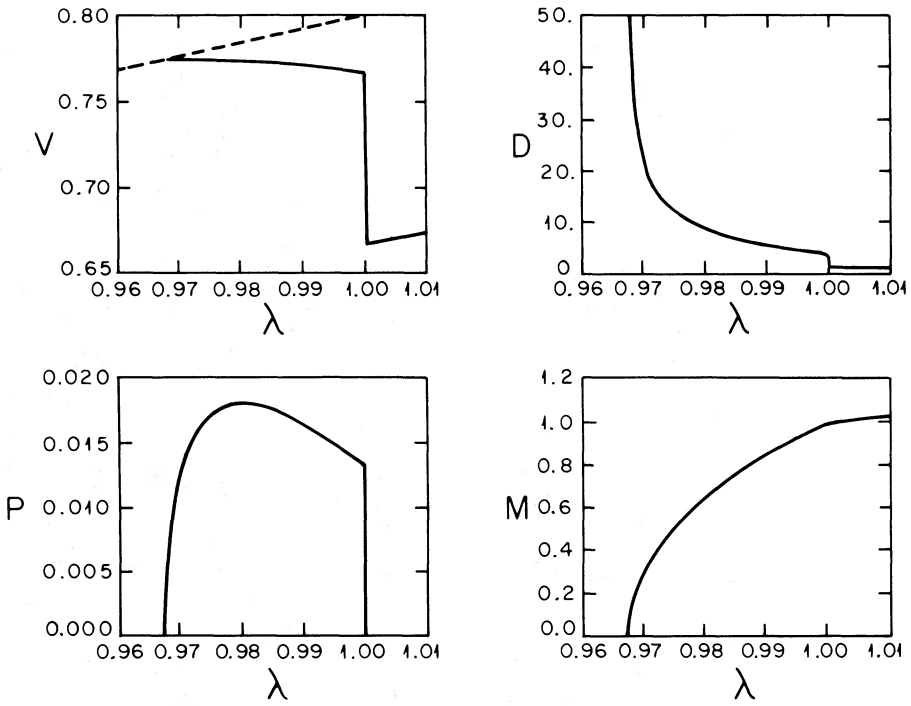


FIG. 1.—Einstein–de Sitter–Sedov solution for $\gamma = 5/3$. See note to Table 1 for units. The density is zero inside $\lambda = 0.9674$. The dashed line in the velocity graph is $V = \delta\lambda$, the velocity a fluid element would have to have in order to remain at constant λ.

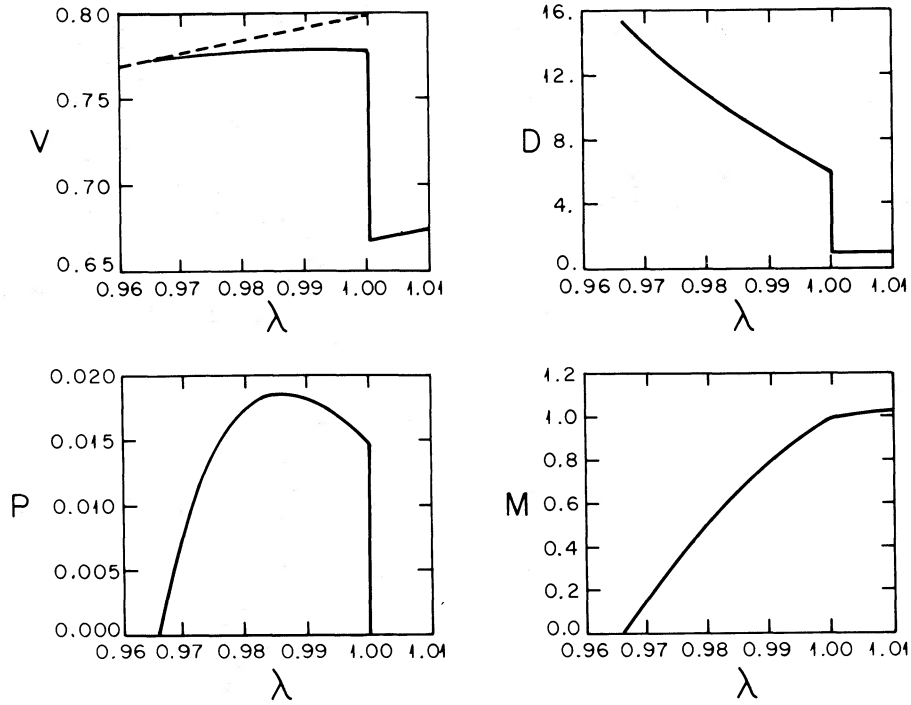


FIG. 2.—Einstein–de Sitter–Sedov solution for $\gamma = 7/5$. See Fig. 1 legend for units and the dashed line. The density is zero inside $\lambda = 0.9661$.

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TABLE 1

EINSTEIN–DE SITTER–SEDOV SOLUTION FOR $\gamma = 5/3$[a]

λ	V	D	P	M
1.000000	0.766667	4.00000	0.0133333	1.000000
0.998816	0.767386	4.15942	0.0137546	0.985523
0.997588	0.768070	4.33174	0.0141804	0.969946
0.996317	0.768720	4.51878	0.0146089	0.953182
0.995003	0.769336	4.72273	0.0150382	0.935131
0.993646	0.769917	4.94625	0.0154654	0.915674
0.992246	0.770463	5.19266	0.0158871	0.894677
0.990801	0.770974	5.46613	0.0162986	0.871979
0.989312	0.771450	5.77196	0.0166943	0.847386
0.987778	0.771889	6.11710	0.0170662	0.820667
0.986199	0.772292	6.51078	0.0174045	0.791538
0.984572	0.772658	6.96563	0.0176954	0.759641
0.982898	0.772985	7.49954	0.0179205	0.724522
0.981175	0.773273	8.13876	0.0180540	0.685582
0.979401	0.773521	8.92396	0.0180590	0.642007
0.977573	0.773725	9.92226	0.0178798	0.592637
0.975688	0.773884	11.25494	0.0174285	0.535717
0.973743	0.773995	13.17161	0.0165529	0.468337
0.971730	0.774051	16.30652	0.0149517	0.384940
0.969639	0.774044	23.10934	0.0118475	0.271592
0.967437	0.773950	∞	0	0

[a] $\lambda = r/r_s$ is the radius in units of the shock radius. V, D, P, and M are the nondimensional velocity, density, pressure, and mass, respectively, with units given by dimensional combinations of the explosion parameters t_i, E, and ρ_i (cf. eqs. [1]–[3]).

TABLE 2

EINSTEIN–DE SITTER–SEDOV SOLUTION FOR $\gamma = 7/5$[a]

λ	V	D	P	M
1.000000	0.777778	6.00000	0.0148148	1.000000
0.998980	0.778073	6.21054	0.0152574	0.981330
0.997910	0.778328	6.43427	0.0157021	0.961111
0.996790	0.778543	6.67240	0.0161449	0.939197
0.995615	0.778714	6.92628	0.0165809	0.915423
0.994384	0.778841	7.19745	0.0170032	0.889605
0.993093	0.778919	7.48762	0.0174035	0.861530
0.991740	0.778947	7.79871	0.0177704	0.830960
0.990319	0.778922	8.13294	0.0180896	0.797623
0.988828	0.778840	8.49277	0.0183424	0.761206
0.987261	0.778698	8.88105	0.0185047	0.721351
0.985614	0.778491	9.30100	0.0185451	0.677648
0.983881	0.778216	9.75633	0.0184232	0.629623
0.982057	0.777868	10.25129	0.0180860	0.576724
0.980134	0.777440	10.79078	0.0174643	0.518315
0.978105	0.776929	11.38047	0.0164676	0.453649
0.975963	0.776326	12.02693	0.0149762	0.381857
0.973698	0.775625	12.73781	0.0128322	0.301914
0.971300	0.774818	13.52201	0.0098252	0.212616
0.968759	0.773896	14.38998	0.0056739	0.112541
0.966063	0.772851	15.35396	0	0

[a] See note to Table 1 for units.

performed using cubic splines interpolation, with an asymptotic power series fit in x and x^s near $x = 0$ (where the derivatives of $\tilde{D}$, $\tilde{P}$, and $\tilde{M}$ are singular). Both the virial theorem equation (27) and the mass equation (11d) were also integrated and checked and had relative errors less than 5×10^{-10}. For $\gamma = 5/3$ these results give $r_s = (6\pi)^{1/5}\alpha(EG)^{1/5}t^{4/5} = 1.89(EG)^{1/5}t^{4/5}$. Schwarz, Ostriker, and Yahil (1975), who first gave the form of this solution, gave the coefficient as 2.09 due to an error in accounting for the energy. We note from Table 3 that since the pressure goes to zero in the shell, thermal energy contributes very little to the total energy, which is dominated by the kinetic component. In the absence of a shock (pure Hubble flow) we would have $t = -w = 2/45$, $u = 0$.

The density singularity appears troubling but actually never occurs in practice. Although the dimensionless density D increases with time at $\lambda \approx \lambda_0$ for $\gamma > 7/5$, the real (dimensional) density ρ actually decreases. This may be seen by considering the Lagrangian time derivative (following a fixed fluid element):

$$\frac{d}{d\tau} = \frac{\partial}{\partial\tau} + \frac{(V - \delta\lambda)}{\tau}\frac{\partial}{\partial\lambda} = \frac{\partial}{\partial\tau} - (\delta - V')\frac{x}{\tau}\frac{\partial}{\partial x}. \quad (31)$$

Now equations (3), (10) and (29) give

$$\frac{d(\ln\rho)}{d(\ln\tau)} \approx -2 - (\delta - V')(s - 1) \quad (32)$$

in the limit $x \to 0$. Since x is monotone increasing in λ, $V' = dV/d\lambda < \delta$. Furthermore, $s > \frac{1}{6}$, so $d(\ln\rho)/d(\ln\tau) < -4/3$ and the density of a fluid element at $\lambda \approx \lambda_0$ decreases with time. The shock only approaches the singular self-similar form in rescaled variables.

TABLE 3

EINSTEIN–DE SITTER–SEDOV SOLUTION CONSTANTS

Parameter	$\gamma = 5/3$	$\gamma = 3/2$	$\gamma = 7/5$	$\gamma = 4/3$
λ_0	0.967437	0.967248	0.966063	0.963794
V_0	0.773950	0.773799	0.772851	0.771035
D_0	1.45821	4.04520	15.3540	92.8734
P_0	0.564291	1.56508	5.93316	35.8044
M_0	10.2359	28.3842	107.471	647.025
α	1.04901	1.04612	1.04381	1.04186
t/E[a]	1.58979	1.57674	1.56586	1.55633
u/E	0.0119112	0.0155019	0.0189841	0.0223802
w/E	−0.601700	−0.592247	−0.584843	−0.578707

[a] Here $E \equiv t + u + w$.

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c) Initial Development of Shock Solutions

The cosmological shock solutions presented above are (in the absence of instabilities to be discussed below) approached asymptotically only at large times, because the initial conditions of the explosion do not satisfy self-similarity. First, a blast occurs in a finite volume of space and may also be spread in time, e.g., continuous energy injection for some period of time Δt_i after $t = t_i$. The shock wave formed becomes self-similar only after it has swept up several times the initial explosion mass, and $t - t_i \gg \Delta t_i$. Assuming this occurs in less than a Hubble time ($t - t_i \ll t_i$), the shock enters a phase during which it satisfies the Sedov solution. In this case the cosmological density is constant and $v_s \gg v_H = Hr_s$, so the Hubble flow is dynamically unimportant. This non-cosmological phase is also indicated by the invalidity of equation (2) for $\tau = 1$ ($r_s = 0$). Finally, for t greater than several Hubble times ($\tau \gg 1$) cosmology becomes important, and the shock asymptotically satisfies the cosmological solutions given above. The time at which this occurs may be estimated by equating the shock velocity given by the Sedov solution to that given by the cosmological solution:

$$\frac{0.4r_s}{t - t_i} = \frac{0.8r_s}{t} . \tag{33}$$

This gives $t = 2t_i$.

d) Gravitational Instability and Shell Fragmentation

The discussion here is confined to $n = 2/3$, $\gamma = 5/3$, since this is the case that would be expected in practice. We saw above that for $\tau \gg 1$ the dimensionless density increases at $\lambda = \lambda_0$, becoming asymptotically singular, while the pressure (and hence temperature) decrease. For isothermal perturbations (cooling times become short in the dense shell) this is just the condition for the simple Jeans instability to occur, though we must also consider the expansion of the shell in order to determine if the perturbed mass is gravitationally bound.

Consider a cylindrical fluid element at $\lambda \gtrsim \lambda_0$ ($\lambda_0 \lesssim \lambda \lesssim \lambda_0 + \Delta\lambda$) of height $\Delta\lambda$ and radius Δs (lengths are measured in units of r_s), whose axis points in the radial direction. Neglecting the effects of a perturbation on the surrounding medium, the element is unstable to gravitational collapse if its total energy is negative. It may be shown that for $\Delta\lambda \gtrsim \Delta s$ the energy is positive, so we only need consider $\Delta\lambda < \Delta s$. In this case the gravitational energy is given to a good approximation by that of a thin disk, $-(8/3\pi)GM^2/(r_s \Delta s)$. Then the nondimensional kinetic, thermal, and gravitational energies of the fluid element (measured in the moving frame at $\lambda = \lambda_0$) are, to lowest order in $\Delta\lambda$ and Δs,

$$\begin{aligned} t &\approx \frac{9\pi}{20}\delta^{14/9}D_0(\Delta s)^4(\Delta\lambda)^{5/9} \approx 1.46(\Delta s)^4(\Delta\lambda)^{5/9} , \\ u &\approx \tfrac{27}{28}\pi\delta^{5/9}P_0(\Delta s)^2(\Delta\lambda)^{14/9} \approx 1.51(\Delta s)^2(\Delta\lambda)^{14/9} , \\ w &\approx -\tfrac{36}{25}\delta^{-8/9}D_0{}^2(\Delta s)^3(\Delta\lambda)^{10/9} \approx -3.73(\Delta s)^3(\Delta\lambda)^{10/9} . \end{aligned} \tag{34}$$

The kinetic energy is due entirely to the spherical divergence, since the radial velocity gradient is very small near $\lambda = \lambda_0$. (Radial kinetic energy $\lesssim 10^{-4}u$.)

There are now three constraints on the dimensions of

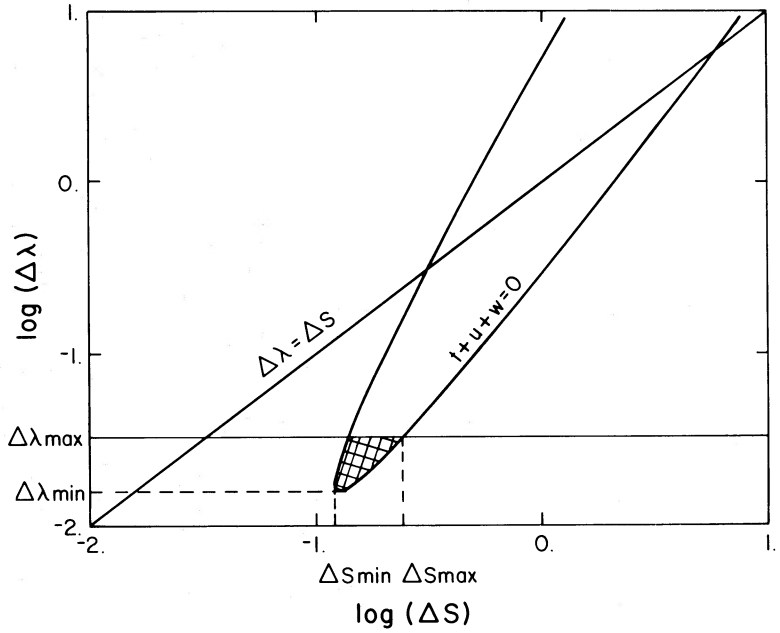


FIG. 3.—Gravitational instability region for the $\gamma = 5/3$ Einstein–de Sitter–Sedov solution. $\Delta\lambda$ and Δs are the height and radius, respectively, of a cylindrical fluid element. Lengths are in units of the shock radius r_s. $(\Delta\lambda_{min}, \Delta\lambda_{max}, \Delta s_{min}, \Delta s_{max}) = (0.01592, 0.03256, 0.1285, 0.2440)$. The total energy (kinetic + thermal + gravitational) of the fluid element is $t + u + w$.

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a fluid element in order that it be unstable: $t + u + w < 0$, $\Delta\lambda < \Delta s$, and $\Delta\lambda <$ (shell thickness) $= \Delta\lambda_{\max}$. These three conditions define a region in the $(\Delta s, \Delta\lambda)$-plane shown in Figure 3. It is seen that fluid elements are unstable if $0.016 \lesssim \Delta\lambda \lesssim 0.033$ and $0.13 \lesssim \Delta s \lesssim 0.24$. The smallest unstable element has mass $\sim 1/300$ of the swept-up mass. The lower limit on Δs for instability corresponds to a maximum of $l \approx 23$ for nonradial perturbations expanded in spherical harmonics Y_{lm}, comparable with the result obtained by Vishniac (1982), who used the thin shell approximation for adiabatic perturbations. Vishniac also showed that when the interaction between the shell and the perturbed element is included, elements are unstable for $\Delta s > 0.24$, corresponding to a minimum $l = 1$.

Thus there is presumably a range of unstable wavelengths for both adiabatic and isothermal perturbations. When the gravitational instability first occurs ($\tau \gtrsim 2$), there will be a minimum unstable mass, which may set the mass scale for galaxies. Furthermore, the positive pressure gradient and negative density gradient at $\lambda = \lambda_0$ cause a lighter fluid element to decelerate a denser one, so a Rayleigh-Taylor instability may also appear (Spitzer, private communication), although this is complicated by the noninertial shock frame. The details of instabilities depend on the transition of the shock from the Sedov to the cosmological phase, which is beyond the scope of this paper.

IV. APPLICATION TO GALAXY FORMATION

In this section the self-similar shock solutions derived above are applied to the explosive amplification model of galaxy formation of OC. In this picture supernova remnants in a young galaxy overlap to produce a galactic wind which leaves the galaxy and forms a shock wave (galactic shock) propagating into the intergalactic medium. The shock forms a dense shell, as a result of cosmological expansion and/or gas cooling, which is gravitationally unstable and collapses to form new galaxies. If the mass swept up by the shock exceeds the initial galaxy mass, amplification results and a chain reaction may occur. The model predicts that (1) galaxy formation occurred recently ($z \lesssim 10$); (2) the diffuse X-ray background (Boldt 1978) may be due in part to thermal bremsstrahlung emission from the galaxy explosions (Schwarz, Ostriker, and Yahil 1975; Bookbinder *et al.* 1980); (3) the cooling shells may produce observable redshifted Lyα and He II λ304 emission (Shull and Silk 1981); and (4) if explosive amplification is sufficiently prolific, the shocks overlap to produce a low-density hot ionized intergalactic medium in pressure equilibrium with embedded cool clouds (Field 1972; OC), much like the interstellar medium (Cox and Smith 1974; McKee and Ostriker 1977). For convenience in this discussion we take $\Omega_0 = (\rho/\rho_{\rm crit})_0 = 1$ and assume that before galaxy formation occurs all the mass is in the pregalactic medium, with $n(\mathrm{He}) = 0.10n(\mathrm{H})$. (The possibility of massive neutrinos is neglected.) The results will not change much if $\Omega_0 \approx 0.5$, as galaxy formation is expected to occur before $z = 1$, in which case $\Omega \approx 1$ at that epoch. We first neglect gas cooling and then consider what constraints it imposes on galaxy amplification. The possibly observable consequences of inverse Compton cooling are also noted.

The tentative nature of the discussion must be emphasized since analytic similarity solutions are not strictly applicable, partly because cooling effects are not well accounted for. A more detailed treatment awaits numerical simulations (Vishniac, Ostriker, and Bertschinger 1983).

a) *Adiabatic Galactic Shocks*

Assume a typical exploding young galaxy has mass $10^{11}M_{i11}\,M_\odot$. Assuming an energy conversion efficiency $10^{-4}\epsilon_{-4}$, the energy produced in a galactic explosion is $E = 10^{61}E_{61}$ ergs $= \epsilon M_i c^2$, or

$$E_{61} = 1.79\epsilon_{-4}M_{i11}\,. \tag{35}$$

This is consistent with the argument of Bookbinder *et al.* (1980) that supernovae must have contributed $\sim 10^{61}$ ergs per galaxy during formation in order to produce the abundance of heavy elements observed in Population II objects. The energy is injected continuously beginning at a redshift z_i over a period Δt_i, the dynamical free-fall collapse time of a protogalaxy. The cosmic time (measured from the big bang) and density at the epoch z_i are

$$t_i = \frac{2}{3H_0}(1+z_i)^{-3/2} = 6.52\times 10^9 h_0{}^{-1}(1+z_i)^{-3/2}\ \mathrm{yr}\,,$$
$$\rho_i = \frac{3H_0{}^2}{8\pi G}(1+z_i)^3 = 1.88\times 10^{-29}h_0{}^2(1+z_i)^3\ \mathrm{g\ cm^{-3}}\,, \tag{36}$$

where h_0 is the Hubble constant H_0 divided by 100 km s^{-1} Mpc^{-1}. Assuming a collapsing protogalaxy cools rapidly and is initially homogeneous, with a density $\rho = D_i\rho_i$, the dynamical time is

$$\Delta t_i = t_{\rm dyn} = \left(\frac{3\pi}{32G\rho}\right)^{1/2} = \frac{3\pi}{4}D_i{}^{-1/2}t_i\,. \tag{37}$$

Since $\Delta t_i \sim t_i$, the shock never enters a Sedov phase. For $t - t_i < \Delta t_i$, supernovae inject energy continuously with $E = E(t) \propto Z(t) =$ galactic metal abundance (cf. Audouze and Tinsley 1976). During the early galaxy collapse $E(t) \propto \tau^\nu$, with $\nu \approx 1$ [$\tau \equiv (t - t_i)/t_i$ during this injection phase]. Then it is clear from dimensional arguments that a self-similar shock wave forms given approximately by the Sedov solution, except with E being explicitly time-dependent. The shock radius, velocity, and swept-up mass are (to within factors of order unity) just the same as for the corresponding Sedov solution with time-varying energy. The internal structure of the shock will differ from the Sedov solution but will be characterized by decreasing density and increasing temperature for $\lambda = r/r_s < 1$. (These also imply constant pressure as $r \to 0$, since the sound speed becomes much greater than the shock speed and pressure gradients are eliminated

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quickly by sound waves.) Since the shock solution is necessary in this phase to determine under what conditions gas cooling is important, before $\tau \sim 1$ (when cosmological effects become important) we assume that it is given by the Sedov solution, which has shock radius, velocity, mass, and interior temperature

$$r_s = 1.15\left(\frac{E(t-t_i)^2}{\rho_i}\right)^{1/5} = 2.77\tau^{2/5}E_{61}{}^{1/5}h_0{}^{-4/5}(1+z_i)^{-6/5}\ \mathrm{Mpc}\ , \quad (38a)$$

$$v_s = \frac{2}{5}\frac{r_s}{t-t_i} = 166\tau^{-3/5}E_{61}{}^{1/5}h_0{}^{1/5}(1+z_i)^{3/10}\ \mathrm{km\ s^{-1}}\ , \quad (38b)$$

$$M_s = \tfrac{4}{3}\pi r_s{}^3\rho_i = 2.47\times 10^{13}\tau^{6/5}E_{61}{}^{3/5}h_0{}^{-2/5}(1+z_i)^{-3/5}\ M_\odot\ , \quad (38c)$$

$$T = \frac{P}{D}\frac{\mu}{k}\left(\frac{r_s}{t-t_i}\right)^2 = 1.27\times 10^7\frac{P}{D}\tau^{-6/5}E_{61}{}^{2/5}h_0{}^{2/5}(1+z_i)^{3/5}\ \mathrm{K}\ , \quad (38d)$$

where k is Boltzmann's constant, H and He are assumed fully ionized so that μ = mean mass per particle = $0.61m_{\rm H}$, and P and D are the dimensionless pressure and density. It is to be understood that $E_{61}(\tau) \approx \tau^{\nu}$, so that, e.g., $r_s \propto \tau^{(2+\nu)/5}$. The coefficient of 1.15 in equation (38a), correct for the $\nu = 0$ Sedov solution, has a slight dependence on ν which is given by Ostriker, McKee, and Cowie (1982), but which will not change the results below, since the ν-dependence may be absorbed into E_{61} or ϵ_{-4}. (Weaver *et al.* 1977 give the coefficient as 0.88 for the case $\nu = 1$.) Shock jump conditions give $P/D = 0.03$ ($T \propto 3.82\times 10^5$ K) at $\lambda = 1$. In the Sedov solution $P/D \propto \lambda^{-\eta}$ with $\eta = 9/2$ for $\lambda \ll 1$ (cf. Sedov 1959); η may be different (but positive) in the injection modified solution.

If gas cooling is negligible, the injection phase lasts until $t - t_i \approx \Delta t_i$, when energy injection ends and cosmological effects become important. A cosmological shock wave forms with

$$r_s = 1.05\left(\frac{Et_i{}^2}{\rho_i}\right)^{1/5}\tau^{4/5} = 2.52\tau^{4/5}E_{61}{}^{1/5}h_0{}^{-4/5}(1+z_i)^{-6/5}\ \mathrm{Mpc}\ , \quad (39a)$$

$$v_s = \frac{4}{5}\frac{r_s}{t} = 303\tau^{-1/5}E_{61}{}^{1/5}h_0{}^{1/5}(1+z_i)^{3/10}\ \mathrm{km\ s^{-1}}\ , \quad (39b)$$

$$M_s = \tfrac{4}{3}\pi r_s{}^3\rho_i\tau^{-2} = 1.87\times 10^{13}\tau^{2/5}E_{61}{}^{3/5}h_0{}^{-2/5}(1+z_i)^{-3/5}\ M_\odot\ , \quad (39c)$$

$$T = \frac{P}{D}\frac{\mu}{k}\frac{r_s{}^2}{t^2} = 1.16\times 10^7\frac{P}{D}\tau^{-2/5}E_{61}{}^{2/5}h_0{}^{2/5}(1+z_i)^{3/5}\ \mathrm{K}\ . \quad (39d)$$

Here $\tau \equiv t/t_i$, $P/D = 1/300$ ($T \propto 3.86\times 10^4$ K) at $\lambda = 1$, and H only is ionized by the shock so $\mu = 0.67m_{\rm H}$. Because of the Hubble flow the velocity of the external fluid with respect to the shock is only $v_s - v_{\rm H} = (\frac{1}{6})v_s$. (This is why the postshock temperature is substantially lower than in the Sedov case.) The density rises and temperature falls behind the shock, with the density nearly zero inside $r_0 = 0.967r_s$ (cf. § III for details).

The swept-up matter is confined to a thin shell of thickness

$$\Delta r_s = 82\tau^{4/5}E_{61}{}^{1/5}h_0{}^{-4/5}(1+z_i)^{-6/5}\ \mathrm{kpc}\ , \quad (40)$$

which is close to galactic dimensions. Interestingly, this scale length is not directly dependent on gas cooling processes, which have been postulated to account for galaxy sizes by Rees and Ostriker (1975) and Silk (1977). (This is not to argue that radiative processes may not be important in determining the mass of a collapsing protogalaxy after a dense shell forms, only to point out that cosmology may also be relevant.)

The shell is unstable to gravitational fragmentation and collapse, which occur adiabatically on a Hubble time scale t_i or isothermally on a cooling time scale $t_{\rm cool}$ (cf. § III*d*). If fragmentation occurs at $\tau = \tau_f \gtrsim 2$, the mass swept up in the shell is (cf. eq. [39c])

$$M_s = 2.65\times 10^{13}\tau_f{}^{2/5}\epsilon_{-4}{}^{3/5}M_{i11}{}^{3/5}h_0{}^{-2/5}\times(1+z_i)^{-3/5}\ M_\odot\ , \quad (41)$$

where equation (35) has been used for E_{61}. Large amplification occurs for initial $10^{11}\ M_\odot$ galaxies. Amplification also occurs on other mass scales, but much smaller masses may be Jeans stable, as will be shown below.

Explosive amplification in the cosmological phase is limited to a moderate range of redshifts z_i. For small z_i there is not enough time for shell fragmentation to occur before the present epoch; this implies $1 + z_i > \tau_f{}^{2/3}$. For $z_i \gtrsim 8$, inverse Compton and free-free cooling occur in less than a Hubble time and cool galactic shocks while still in the energy injection phase. These effects are discussed below.

b) Gas Cooling

Gas cooling is important since it (1) restricts the range of redshifts for which explosive amplification of galaxies may occur and (2) sets the scale of galaxy masses in fragmenting dense shells (the Jeans mass at 10^4 K). Nevertheless, it will be shown that because the energy injection time scale is roughly the Hubble time, the adiabatic cosmological shock solution correctly gives the radius and mass of a dense shell. Thus both cosmological and gas cooling processes are important to galaxy formation in the explosive amplification model.

First consider cooling in the energy injection phase,

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$0 \le \tau \lesssim \Delta t_i/t_i = (3\pi/4)D_i^{-1/2}$. There are three main gas cooling processes: inverse Compton scattering, thermal bremsstrahlung, and line emission. For each, a cooling time $t_{\rm cool}$ for the shock may be calculated. Cooling is dynamically important in this phase if $t_{\rm cool} < t - t_i = \tau t_i$. But in this case individual supernova remnants within a galaxy cool before overlapping to power a galactic wind, so galactic-scale shocks cannot form. Alternatively, a lower mass cooled dense shell forms and fragments, with mass given by equation (38c), with $E_{61} \approx \tau^\nu < 1$. Amplification of galaxy-size masses cannot occur for $t_{\rm cool} \ll \Delta t_i$, setting an upper limit on the redshift as shown below.

At large redshifts, cooling by inverse Compton (IC) scattering of electrons off the cosmic background radiation dominates shock evolution. For a current background temperature of $T_{\rm bkg} = 3$ K the cooling time (of electrons and protons) for IC scattering is (Rybicki and Lightman 1979)

$$t_{\rm IC} = \frac{3}{4}\frac{m_e c}{\sigma_T a T_{\rm bkg}^4} = 1.59 \times 10^{12}(1+z)^{-4}\ {\rm yr}\ . \quad (42)$$

From the above, IC cooling prevents galaxy amplification if $t_{\rm IC} < \tau t_i$, giving

$$1 + z_i > (1 + z_{\rm IC})\tau^{-2/5} \equiv 9.0 h_0^{2/5}\tau^{-2/5}\ . \quad (43)$$

This result was also given by OC (except for the explicit τ-dependence, included here since we are comparing $t_{\rm IC}$ with $t - t_i$, not t_i), and is independent of shock structure. Since the injection phase ends at $\tau \approx \Delta t_i/t_i$, IC cooling limits galaxy amplification to redshifts

$$1 + z_i < 6.4 D_i^{1/5} h_0^{2/5}\ . \quad (44)$$

Note that this constraint depends only weakly on the compression D_i of a collapsing protogalaxy. D_i is not expected to be very large (e.g., $\gg 10$), since the shell from which the protogalaxy forms becomes Jeans unstable on galaxy mass scales for $D_i \sim$ a few (cf. § III*d* and eq. [58]). Thus a reasonable upper limit on the beginning of explosive amplification of galaxy-size objects may be $z \approx 10$.

If galaxy-size explosions do occur as early as $z \approx z_{\rm IC}$, the IC cooling may be observable. IC scattering shifts cosmic background photons to higher energies, producing a dip in the microwave brightness temperature (Zel'dovich and Sunyaev 1969), which has been observed in the direction of nearby ($z < 0.2$) rich clusters of galaxies (Lake and Partridge 1977; Birkinshaw, Gull, and Northover 1978). The magnitude of the microwave dip is (Sunyaev 1980)

$$\frac{\Delta T_b}{T_b} = -2\int \frac{kT_e}{m_e c^2}\sigma_T n_e\, dl \quad (45)$$

integrated along the line of sight, where n_e and T_e are the electron number density and temperature, and l is the path length of the scattering gas. The coefficient of 2 is approximately correct for $h\nu/kT_{\rm bkg} \lesssim 0.1$. Looking face on to a Sedov shock at $\tau = 1$ gives the rough estimate

$$\frac{\Delta T_b}{T_b} \approx -2\int_0^1 P d\lambda \frac{\rho_i r_i^3}{m_e}\frac{\sigma_T}{c^2 t_i^2}$$
$$\approx 4.5 \times 10^{-6} E_{61}^{3/5} h_0^{64/25}\left(\frac{1+z_i}{1+z_{\rm IC}}\right)^{12/5}\ . \quad (46)$$

Since $n_e T_e$ rises out to the shock, this estimate may be increased by several times if looking edge-on to a shock (limb brightening). Since the current detection limit is $\Delta T_b/T_b \sim 10^{-4}$, if IC cooling galactic shocks exist, they may be just visible at $1 + z \approx 9$ as lower temperature rings in the microwave background. The rings would have angular radius (Weinberg 1972) $(27/2)(H_0 r_s/2c) \approx 1' E_{61}^{1/5} h_0^{1/5}$.

Radiative processes may also limit explosive galaxy amplification by cooling galactic shocks in the injection phase. For temperatures above $\sim 10^6$ K, free-free emission (thermal bremsstrahlung) is the dominant radiative cooling process. For a hydrogen plasma the free-free cooling function is (Spitzer 1978, p. 58, evaluating $g_{\rm ff}$ at 10^6 K)

$$\Lambda_{\rm ff} = 2.01 \times 10^{-27} T^{1/2}\ {\rm ergs\ cm^3\ s^{-1}}\ . \quad (47)$$

This leads to a cooling time (including 10% He by number) in the injection phase of

$$t_{\rm ff} = \frac{1.5p}{1.4\Lambda_{\rm ff} n_e n_{\rm H}} = 1.99 \times 10^{12}(P/D^3)^{1/2}$$
$$\times \tau^{-3/5} E_{61}^{1/5} h_0^{-9/5}(1+z_i)^{-27/10}\ {\rm yr}\ . \quad (48)$$

At $\lambda = 1$, $(P/D^3)^{1/2} = 0.0433$ and $(P/D^3)^{1/2} \propto \lambda^{-3\eta/2}$ for $\lambda \ll 1$. Free-free cooling occurs if $t_{\rm ff} < \tau t_i$, giving

$$1 + z_i > 8.6\tau^{-4/3} E_{61}^{1/6} h_0^{-2/3} \quad (49)$$

at $\lambda = 1$; the numerical coefficient increases as $\lambda^{-5\eta/4}$ inside the shock and is only approximate since we have assumed the Sedov solution in the injection phase. Since $E_{61} \approx \tau^\nu$, equation (49) provides an upper limit to the redshifts for which galaxy amplification may occur if $\nu \le 8$, which is almost certainly the case. For $\tau < \Delta t_i/t_i$ and $\lambda \le 1$ the limit is

$$1 + z_i < 2.7 D_i^{2/3} E_{61}^{1/6} h_0^{-2/3}\ . \quad (50)$$

Since we expect $D_i \sim 5$–10 ($D_i \ge 4$ from the shock jump conditions), this limit is comparable with that given by IC cooling in equation (44) for $E_{61} \sim 1$; the dependence on E_{61} prevents the amplification of very low mass galaxies.

Below 10^6 K, radiative cooling at the epoch of galaxy formation is due almost entirely to Lyα and He II λ304 emission. Shull and Silk (1981) have calculated the emission from a 100 km s^{-1} shock using the radiative shock code of Shull and McKee (1979). While their shock model is not strictly valid for the injection-modified Sedov solution, the results should not be very sensitive to internal structure, since virtually all the radiation comes from near the shock (where the density is highest).

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Thus their calculated luminosity should also be valid here, provided $E_{61} \approx \tau^{\nu}$. For metals depleted to 10^{-3} of solar abundances the Lyα luminosity is (assuming line intensity $\propto v_s^2$, as holds approximately in Shull and Silk 1979, Fig. 1)

$$L_{\mathrm{Ly}\alpha} = 8.11 \times 10^{-11} \left(\frac{v_s}{100 \text{ km s}^{-1}}\right) 4\pi r_s^2 n_{\mathrm{H}} v_s$$
$$= 4.31 \times 10^9 \tau^{-2/5} {E_{61}}^{4/5} {h_0}^{4/5} (1 + z_i)^{6/5} L_{\odot} . \quad (51)$$

(The line intensity is a factor of 4 greater than the value given by Shull and Silk 1981 due to textual errors in their paper [Shull, private communication]. The luminosity given here agrees with theirs, $6 \times 10^8 L_{\odot}$ at $\tau = 1$ for $E_{61} = 0.1$.) Since He II λ304 has the same luminosity and all other lines contribute less than 1%, the cooling time for the entire shock due to line emission is

$$t_l = \frac{U}{2L_{\mathrm{Ly}\alpha}} = \frac{0.717E}{2L_{\mathrm{Ly}\alpha}}$$
$$= 6.89 \times 10^9 \tau^{2/5} {E_{61}}^{1/5} {h_0}^{-4/5} (1 + z_i)^{-6/5} \text{ yr} . \quad (52)$$

Line cooling occurs in the injection phase if $t_l < \tau t_i$, giving

$$1 + z_i < 0.8 \tau^2 {E_{61}}^{-2/3} {h_0}^{-2/3} . \quad (53)$$

If $\nu \leq 3$, line cooling thus imposes no constraint on galaxy amplification. If $\nu > 3$, initial supernovae cool, but for larger τ, as the star formation rate increases, remnants succeed in overlapping before line cooling occurs, so it is again ignorable.

OC found radiative cooling to be effective for $z \lesssim 5$, but they assumed solar cosmic abundances in calculating the cooling. Metal abundances in the galaxy formation epoch certainly would not be expected to have exceeded those of Population II objects, $\sim 10^{-2}$ of solar. Since radiative cooling for $10^5 \text{ K} < T < 10^6$ K and solar abundances comes primarily from O III–VI line emission (Cox and Tucker 1969), the cooling time is longer than OC give and, as shown above, radiative cooling is probably not important in leading to shell formation.

Thus for $z_i \lesssim z_{\mathrm{IC}}$ gas cooling is unimportant during the energy injection phase, allowing galaxy-size explosions to occur and propagate into the intergalactic medium. But since the injection phase ends at $t - t_i \approx t_i$ (protogalaxy collapse time ~ Hubble time), cosmological effects promptly become important. The shock cools adiabatically and develops a thin dense shell (cf. § III*b*), propagating according to equations (39abc). Once this occurs, however, radiative processes take over and dominate the shell fragmentation. This may be seen from the result corresponding to equation (53) for adiabatic cosmological shocks ($v_s \to v_s - v_{\mathrm{H}}$, $U = 0.0119E$). Radiative cooling dominates in the dense shell ($t_l < t_i$) for

$$1 + z_i < 13 f^{10/3} \tau^{-8/3} {E_{61}}^{-2/3} {h_0}^{-2/3} , \quad (54)$$

where f is the factor by which cooling is increased over that calculated from the Shull and Silk (1981) model assuming only Lyα cooling. The explicit f-dependence is given because the calculations by Shull and Silk do not correctly account for the self-gravity of the shell, which increases shell densities and hence cooling rates in the cosmological phase, so $f > 1$. If $v_s - v_{\mathrm{H}} > 100$ km s^{-1}, He II λ304 emission is important, which doubles f.

Even though radiative cooling occurs, cosmological effects still govern the shell propagation (and hence mass), since, as Ostriker, McKee, and Cowie (1982) have shown, for $t - t_i > t_i$, cooled shocks propagate as $r_s \propto t^{\delta}$, with

$$\delta = \frac{15 + (17)^{1/2}}{24} \approx 0.797 , \quad (55)$$

which is so close to $\frac{4}{5}$ that equation (41) for the mass is still correct.

It is conceivable that adiabatic expansion may be so effective in cooling the shell that recombination occurs. In that case radiative cooling would decrease greatly and the shell would presumably fragment adiabatically (cf. § III*d*). Since there is likely to be a residual source of heating (such as quasar photoionization), however, temperatures will stabilize at $\sim 10^4$ K in a few cooling times ($t_{\mathrm{cool}} < t_i$), and the shells will resemble H II regions.

Radiative cooling therefore probably sets the mass scale of the galaxies forming from shell collapse. One can estimate the mass of the galaxies as the Jeans mass which can cool at 10^4 K just at the dynamical free-fall rate (Rees and Ostriker 1975; Silk 1977). The cooling rate at 10,000 K for collisional excitation of H is approximately $5.3 \times 10^{-24} n_e n(\mathrm{H\,I})$ ergs cm^{-3} s^{-1} (Spitzer 1978, p. 140). The Jeans mass is defined as

$$M_{\mathrm{J}} \equiv \left(\frac{\pi k T}{G\mu}\right)^{3/2} \rho^{-1/2} . \quad (56)$$

Equating the local cooling time ($= 1.5p/\Lambda n_e n[\mathrm{H\,I}]$) to the dynamical time (eq. [37]) then gives

$$M_{\mathrm{J}} = 5.3 \times 10^{10} M_{\odot} , \quad (57)$$

where the hydrogen ionization fraction has been taken to be 0.5. Using the cosmic density with a compression factor D and temperature $T = T_4 \times 10^4$ K (again with H half ionized) gives a similar result:

$$M_{\mathrm{J}} = 3.4 \times 10^{10} \tau D^{-1/2} {T_4}^{3/2} {h_0}^{-1} (1 + z_i)^{-3/2} M_{\odot} . \quad (58)$$

c) *Explosive Amplification*

The results of the preceding sections suggest that at least one generation of explosive amplification of galaxies can occur for

$$\tau_f^{2/3} \lesssim 1 + z_i \lesssim \min \{6.4 D_i^{1/5} {h_0}^{2/5} ,$$
$$2.7 D_i^{2/3} {E_{61}}^{1/6} {h_0}^{-2/3} , \quad 13 f^{10/3} \tau_f^{-8/3} {E_{61}}^{-2/3} {h_0}^{-2/3} \} , \quad (59)$$

where $f \gtrsim 1$, $D_i \gtrsim 4$, and $\tau_f \gtrsim 2$. The lower limit for z_i is set by the fact that for $1 + z_i < \tau_f^{2/3} \gtrsim 2$ galaxies would still be forming in the present epoch. At high redshifts, inverse Compton and free-free cooling prevent supernova remnants within a galaxy from overlapping

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to power a galactic wind, or the cosmological dense shells are too hot to cool radiatively. (Adiabatic fragmentation may perhaps occur in the latter case, leading to radiative cooling at high enough densities.) Alternatively, for fixed $z_i < z_{IC}$, galaxy formation will not occur if E is too small (free-free cooling prevents galactic shock formation) or too large (cosmological shocks do not radiatively cool). Jeans mass sized galaxies (eq. [58]) do amplify, however. Assuming the initial exploding galaxy has mass equal to the Jeans mass at $z = z_i$ ($\tau = 1$), equation (54) gives the condition for line cooling to dominate in the shells:

$$19 f^{10/3} \tau_f^{-8/3} D^{1/3} T_4^{-1} \epsilon_{-4}^{-2/3} > 1 . \quad (60)$$

Since this inequality is expected to hold, the shells cool and fragment rapidly, suggesting $\tau_f \approx 2$–3.

Assuming the protogalaxies forming out of the fragmenting shells have mass equal to the Jeans mass at $\tau = \tau_f$, the first-generation explosive amplification factor (cf. eqs. [41] and [58]) is given by

$$A_1 = \frac{M_s}{M_J} = 406 \tau_f^{-3/5} D^{1/5} T_4^{-3/5} \epsilon_{-4}^{3/5} . \quad (61)$$

For $\tau_f = 2$, $D = 10$, this gives an amplification ~ 400.

The formation of galaxies in the shell will produce further energy injection into the intergalactic medium, producing a "supergalactic" shock wave. Shock waves from individual galaxies propagate quickly through the low-density cavity inside the shell, so they all overlap to produce a radial shock with energy (cf. equations [35] and [41])

$$E_{61} = 249 \tau_f^{2/5} D^{-3/10} T_4^{9/10} \epsilon_{-4}^{8/5} h_0^{-1} (1 + z_i)^{-3/2} , \quad (62)$$

and radius, velocity, mass and postshock ($\lambda = 1$) temperature at $\tau = \tau_f$ (obtained by substituting eq. [62] into eq. [39])

$$r_s = 7.61 \tau_f^{22/25} D^{-3/50} T_4^{9/50} \epsilon_{-4}^{8/25} \times h_0^{-1} (1 + z_i)^{-3/2} \text{ Mpc} , \quad (63a)$$

$$v_s = 913 \tau_f^{-3/25} D^{-3/50} T_4^{9/50} \epsilon_{-4}^{8/25} \text{ km s}^{-1} , \quad (63b)$$

$$M_s = 5.12 \times 10^{14} \tau_f^{16/25} D^{-9/50} T_4^{27/50} \epsilon_{-4}^{24/25} \times h_0^{-1} (1 + z_i)^{-3/2} M_\odot , \quad (63c)$$

$$T = 3.51 \times 10^5 \tau_f^{-6/25} D^{-3/25} T_4^{9/25} \epsilon_{-4}^{16/25} \text{ K} . \quad (63d)$$

Line cooling dominates in the dense shell if $t_l < t_i$, or (cf. eq. [60]) if

$$0.34 f^{10/3} \tau_f^{-44/15} D^{1/5} T_4^{-3/5} \epsilon_{-4}^{-16/25} > 1 . \quad (64)$$

This inequality may hold if $f \gtrsim 2.2$ for $\tau_f \approx 2$ and $D = 10$, suggesting that line cooling may be marginally effective in contributing to shell fragmentation. (He II $\lambda 304$ emission would give $f \gtrsim 2$.) If radiative cooling does predominate, the shell will fragment into Jeans mass size protogalaxies, with a total amplification (cf. eqs. [58] and [63c])

$$A_2 = 1.5 \times 10^4 \tau_f^{-9/25} D^{8/25} T_4^{-24/25} \epsilon_{-4}^{24/25} . \quad (65)$$

If $\tau_f \approx 2$, there is just enough time for two generations of explosive amplification of galaxies between $1 + z_i = 1 + z_{IC} \approx 9$ and $1 + z = 3$, with total amplification $\sim 2 \times 10^4$, though as mentioned above it is not clear that a second generation can cool. It is interesting to note that a single generation produces $\sim 3 \times 10^{11} M_\odot$ in galaxies, typical of small groups, while two generations produce $\sim 2 \times 10^{13} M_\odot$, characteristic of small clusters. The resultant radii and velocity dispersions are also typical of groups and clusters (eqs. [39] and [63], with $z_i = 8$, $\tau_f = 2, D = 10, T_4 = \epsilon_{-4} = 1$): $r_s \approx 100$ kpc, $v_s \approx 200$ km s^{-1} for groups, and $r_s \approx 500$ kpc, $v_s \approx 700$ km s^{-1} for clusters. Since only a small fraction of the initial explosion energy is radiated away ($u/E \approx 0.01$, cf. § III*b*), the groups and clusters formed would not be gravitationally bound.

V. SUMMARY

Similarity solutions have been found which are valid for times long compared to the Hubble time for adiabatic shock waves in $\Omega \ll 1$ (open) and $\Omega = 1$ (Einstein–de Sitter) Friedmann universes. Dimensional analysis shows that these are the only cases (besides the special Sedov case) for which self-similar shocks exist for homogeneous, self-gravitating fluids. Integrals of the motion were derived expressing conservation of mass and energy, and the adiabatic property of the postshock flow (eqs. [16], [18], and [24]). Even with three integrals (for four fluid variables), however, for the $\Omega = 1$ case only numerical solutions were found.

The $\Omega \ll 1$ case is trivial. There is a contact discontinuity (a density jump) comoving with the Hubble flow, but since no matter passes through it, no shock heating occurs, and the interior is pressureless. The density profile can only be determined from initial conditions, which may vary depending on the value of τ at the time of the explosion.

The $\Omega = 1$ case is more interesting. An initial Sedov shock wave (valid for $t - t_i \ll t_i$) decelerates until at $t - t_i \approx t_i$ the shock velocity is only 6/5 times the Hubble velocity. The shock heating decreases substantially and the interior cools adiabatically, with the matter piling up behind the shock to form a dense shell of thickness $\sim 0.03 r_s$. For $\gamma > 7/5$ the dimensionless density becomes singular at the inner edge, although the real density decreases with time (because of the cosmological expansion). It is argued that the shell is unstable to adiabatic and isothermal perturbations, though no proof is given.

The $\Omega = 1, \gamma = 5/3$ solution is applied to the explosive amplification model of galaxy formation of Ostriker and Cowie (1981). In this model, rapid star formation and supernova explosions occurring in a collapsing protogalaxy power a galactic wind, producing a shock wave propagating into the intergalactic medium. During this energy injection phase a modified Sedov shock wave forms, with time-varying energy. It is shown that for $1 + z \gtrsim 9$, inverse Compton and free-free cooling occur before injection is complete and thus limit galaxy amplification to smaller redshifts (eqs. [44], [50]). Line cooling is unimportant in this phase for low (Population II) metal abundances.

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Energy is injected over the whole collapse time of the galaxy, roughly a Hubble time, so cosmological effects are important in the subsequent shock propagation. The shock evolves to the thin-shelled self-similar form described above. Provided recombination does not occur and the explosion energy is not too great, now line cooling becomes very important and the shell fragments isothermally into new protogalaxies with Jeans mass $\lesssim 10^{11}\ M_\odot$. The mass, radius, and velocity dispersion of the shell are typical of small groups of galaxies. The formation of galaxies in the shell may inject more energy into the intergalactic medium, causing a shock wave to continue propagating radially outward. If line cooling is great enough (this is uncertain), the resulting cosmological shock will fragment to form a second generation of galaxies, with shell mass, radius, and velocity typical of small clusters of galaxies.

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